Emotionally Adaptive AI Companions for Supporting Routine Management in Autistic Adolescents

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***Abstract*—Autistic Adolescents usually experience difficulties in the management of emotions, routine transitions and social cue interpretation. Many existing tools that aim to fill in the gap are often non-personalise, static or lack real-time responsiveness in handling these challenges. This study conceptualises and empirically validates a prototype of an emotionally adaptive AI companion that focuses on reducing stress due to routine transition, emotional regulation and social cue interpretation while increasing personalised management by providing contextual support. A quasi-experimental, mixed methods design is adopted. The core of this system conducts facial multimodal emotion recognition through facial expression and simulated voice tone using transfer learning across three CNN architectures (ResNet-18, MobileNetV2, and EfficientNet-B0) as comparison tests. The resulting emotion output is feeds into a contextual engine for real-time personalised interventions which can also be continuously improved through critical feedback-in-the-loop control architecture based on caregiver logs. The key model trade-offs are validated, the findings established that ResNet-18 possesses the highest accuracy of 48%, EfficientNet-B0 with a superior F1 Score of 0.31 and MobileNetV2 proves to be efficient but slightly lower performance compared to other architectures. Simulated user feedback validation resulted in high preliminary acceptability, as high as 87.5% acceptability for an intervention like "Reassurance". This validated the utility of this responsive system. This transfer-learning based, multi-modal pipeline is robust. The results of the comparative analysis uncovered a very profound and instructive trade-off between the complexity of models, their efficiency, and performance metrics relating to accuracy versus the F1-score.**

***Index Terms*—Affective computing, autism disorder, context-aware systems, emotion recognition, human-in-the-loop, multi-modal sensing, reinforcement learning, transition support, Addaptive system.**

I. Introduction

Individuals with autism often face challenges with routine transitions, social cue interpretation, and emotional regulation. While many support tools exist, they are often static, non-personalised, or lack real-time responsiveness to emotional state. Hence, the need to develop and evaluate a context-aware AI-powered digital companion that support autistic adolescents in managing daily transitions using multimodal emotion recognition and personalised routine planning. The multimodal behavioural input would involve using facial expressions to detect stress or anxiety levels

Autistic adolescents frequently face challenges or issues in navigating their daily lives from regulating emotions, interpreting social to even environmental cues. And these difficulties can contribute to a increased rate of anxiety and reduced independence, especially during the adolescence which is a developmental stage characterised by increasing complex social expectations, self-reliance, and academic demands. While the current digital support systems for autistic individuals offer visual schedules, therapeutic prompts or routine planners, most of them are static or are limited in their ability to adapt to contextual emotional changes in real-time. And as such, they often fail to provide the responsive and personalised support needed to meaningfully assist autistic users during transition-related stress. It is evident with recent advances in affective computing and artificial intelligence (AI) to create more personalised digital, empathetic, and responsive supports. By using multimodal behavioural signals including facial expressions, with voice records, and physiological indicators, combined with contextual information like time of day, location, and known

stress triggers, AI systems can now predict the emotional states and proactively recommend coping strategies. However, existing approaches in autism-focused technology rarely integrate emotion recognition with routine support, do they sufficiently account for neurodiverse communication patterns and sensory experiences. A recent review of AI-assistive technologies for this kind of conditions emphasised the potential of AI to support adaptive functioning in everyday environments but also highlights substantial gaps in long-term evidence, contextual specificity, and translation to routine-heavy domains [1]. [2] has also developed emotion-aware technologies for autistic children but these largely remain on discrete tasks rather than continuous, transition-focused support. This study seeks to focus on these gaps by developing and evaluating a context-aware, AI-powered companion with the goal of supporting autistic adolescents in managing daily routines and transitions. The proposed system combines personalised routine planning, multimodal emotion recognition, reinforcement learning in order to offer an adaptive and proctive guidance delivered through a friendly digital interface. Hence, the system learns and evolves to provide specific support that aligns with personal preferences and needs just by integrating a continuous feedback loop from users and caregivers. Above all, this study seeks to increase independence, reduce transition-related stress, in order to empower autistic adolescents to navigate daily life with confidence, offering a foundation for future AI-driven assistive technologies that are neurodiversity-affirming, context-sensitive, and emotionally intelligent. This study seeks to offer three major contribution to the knowledge

- The targets of autistic adolescents as it is still in a development stage with varied features of high transition demands and emotional complexity, addressing a population and use-case often overlooked in existing solutions that focus primarily on younger children.
- Development of an adaptive companion that evolves through the user or caregiver input and behavioural and physiological feedback that can help to promote autonomy rather than reliance on caregivers.
- Introduction a system that is context-aware and with the capability of proactively predicting high-stress transition periods and delivering tiered interventions, from choice-based prompts to guided coping strategies there by enhancing real-world applicability.

Fig 1 is the conceptual framework for this study and it is built on a dynamic and closed-loop architecture that incorporates learning, sensing, action, and interpretation in order to provide emotionally adaptive support for autistic adolescents during daily transitions. So at the centre, the system continuously collects multimodal behavioural signals, like vocal tone, facial expressions, and physiological indicators. In addiction to contextual information like time and activity. These inputs are feed into the system then it interprets the emotional state, anticipates transitions, and selects personalised support strategies. A support interface then delivers proactive prompts and coping tools, while a feedback and learning layer

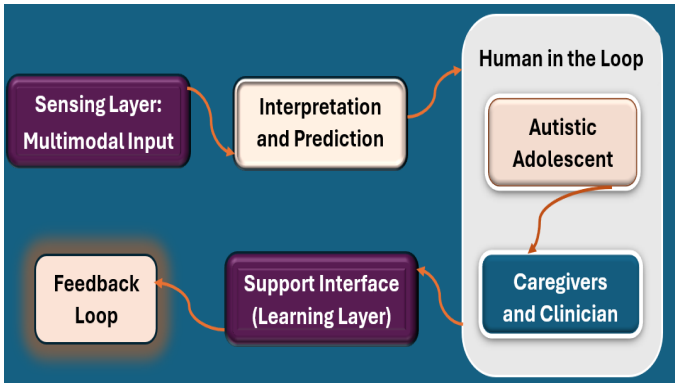

Fig. 1. Autistic Adaptive Companion System(source: Author)

refines decisions over time based on the user and caregiver responses. The system is designed within a human-in-the-loop paradigm, the framework ensures that caregivers and clinicians will remain as informed contributors to the system's adaptive process.

II. LITERATURE REVIEW

A. Multimodal Behavioural input

In recent research, artificial intelligence has shown the ability to enhance adaptive functioning in autism because of its ability to deliver customised daily assistance [1]. Nevertheless, most of the existing tools are fixed, which are unable to react to the dynamic feeling of a user. For instance, parent-managed video prompting can be used to educate on a life skill [3], yet, it incorporates static prompts where the teen is not allowed to be flexible in case of getting anxious or disengaged [4]. To fill this gap, new systems are adding multimodal behavioural input to analyse facial, vocal and physiological inputs in parallel to recognise user emotions more reliably in real time, the subtle or ambiguous signals of either of the single modalities. The influence of multimodal input is demonstrated by the recent demonstrations. [5] , who combined video and biosensor readings in one child-robot companion to constantly assess the interest of the child; they combined face, voice, posture, and heart-rate signals to produce an engagement value, and then the robot modified its behaviour based on the child's condition [5]. Such multi-cue responsiveness is an obvious enhancement of the traditional, scripted techniques. Physiological indicators such as heartbeat can be objectively used to track the increased stress in an autistic person [6] with the use of wearable sensors. [7] used a combination of heart-rate patterns and observed behaviours to estimate which calming intervention would be most appropriate during a stress episode of a student, and their model could trigger a sensory break prior to a complete meltdown, therefore provide support proactively [7]. Scientists are also extending new methods of feeling. [8] describe the application of pre-trained vision and audio models to enhance accuracy in emotion recognition in an AI assistant and demonstrated high performance on standard datasets, indicating that such improvements will enable AI

companions to read between the lines on different expression displays in autism [8].

B. Contextual AI Engine

There is a level of feeling, emotion, understanding the situational context and making choices to act. The contextual artificial intelligence engine combines the data regarding the context of the user, schedule, and habits, to provide assistance at the appropriate moment. The recent research focused on context-sensitive planning for autistic users.[9] suggested a system, which monitors factors such as time of the day, place, and agenda followed by a student to predict stress and change prompts. Prior to an anticipated problematic course, the system provides an early notification and a relaxing activity to help the course changeover feel relaxed [9]. Similarly, [10] introduced AwarNS, the general model of mobile apps to respond to context data in real time, combine triggers and provide corresponding coping advice in real time. Machine learning augments the context engine in terms of its adaptivity. [11] set up deep models with a reinforcement learning agent that executively learnt the optimality of adjustments to the routine that enhanced the daily outcomes of each autistic child, and through successive interactions it attempted various support actions, reinforcing those that resulted in better engagement to personalise the routine plan [11]. Moreso, [7] determined that proactive strategies that the artificial intelligence selected worked better than reactive strategies that were used once the process had gone out of control. This highlights the importance of an engine which is capable of pre-empting and overcoming difficult situations. Widespread tendencies in the field are indicative of this. [12] observed that the emerging new autism support systems are now meant to provide support to people on a continuous context-sensitive support in home and school instead of being limited to single-environment tools. In the meantime, generative AI is opening up opportunities to more natural interactions.[13] demonstrated that an interactive story-based coach with a generative model enhanced the social-emotional learning of autistic children, although experts warn that these large language model-based interventions should be approached with care to make sure they do not get out of hand and serve the interests of users and safety[14].

C. Routine and Transition Support

Among the primary roles of the AI companion is to assist in organising everyday patterns and facilitating the process of switching activities in areas that can be quite difficult in the case of autistic adolescents. Autistic adults engaged deeply with a conversational virtual human and often valued its predictable, judgment-free interaction[15], A program that coached autistic adolescents in daily living skills achieved significant gains in routine independence[16]. Traditional tools such as visual schedules or step-by-step videos may be helpful but will frequently have to involve close caregiver supervision and adjustment in real-time[3]. [17] presented a mobile app named DailyBuddy that allows autistic people to practice various life skills with personalised instructions to compare the various forms of instruction and monitor the performance of the user to determine which type of help is most effective and provides personalised future prompts. Such one-on-one routine coaching resulted in much better independent task completion than a standard one-size-fits-all approach [17]. [18] constructed a virtual campus rehearsal transition where autistic children could engage in routines, such as the transfer between the classrooms, or even going through a cafeteria, with no danger. Within several weeks of practicing in the VR environment, children demonstrated significant progress in executing tasks in the real world, as well as in feeling less anxious about such experiences, which is why immersive, context-focused rehearsal and AI support can help instil better confidence in real-world transits[19]. The insights into designing AI are established by human-delivered interventions. A program of structured group therapy coaching autistic teens in the day-to-day living skills brought about high returns in routine-independence[17]. Although the program did not rely on AI, its effectiveness in a controlled environment implies that an AI may mimic some coaching roles such as reminding a teenager (and their parent) to practice a new skill at home and then provide feedback on their progress [16]. Assistive technology trials also indicate the advantages of context fit and personalization.[19] discovered that systems that were tested with familiar environments where caregivers were involved resulted in increased engagement with autistic users. It means that an AI-based routine support should be tailored to the daily situation of a particular user and, possibly, incorporated in the process of looping caregivers or teachers in when needed.

D. Feedback and Learning Loop

The AI companion needs to adapt the support strategies and keep learning about the user to adjust to conditions, creating feedback and learning loop to stay effective over time. Among the mechanisms is reinforcement learning, in which the system revises its behaviour according to which actions resulted in positive outcomes[11] . In addition to the algorithmic learning, human feedback is vital. The companion seeks the user's input following an intervention to determine whether they have been helpful. The AI uses the input as feedback to attempt a different strategy when a user repeatedly fails to perform a task, a strategy that DailyBuddy has shown to be successful at switching between instructional strategies[17]. During assessments, improved engagement and results were observed in systems where caregivers and educators were allowed to make their input [19]. The companion also understands the preferred behavioural frequency, tone of communication that most effectively soothes the user, and what motivates the user. Apart from enhancing efficacy, alignment with user preferences enhances trust[14]. These explain the reasons why the AI is making such suggestions in a simple manner, which prevents confusion or mistrust[20]. [21] report that autistic participants envisioned AI for coping with negative self-talk that offers judgment-free emotional support. Lastly, skill outcomes can be directly enhanced with the help of closed-loop feedback. A gamified AI tutor responded to the real-time emotional state when modifying its feedback, which held the child more engaged

and enhanced the outcome of emotion-learning measures than static feedback [22]. Within the framework of the routine support, an emotionally adjustive companion that constantly adjusts its instructions can also provide stronger improvements as it becomes more sensitive towards the needs of the user.

III. METHODOLOGY

A. Study Design

This study uses a quasi-experimental approach integrated with the help of mixed methods toward the design and evaluation of a new, emotionally adaptive Artificial Intelligence companion for the support of routine management in adolescents with autism. Its broader goals are to develop an intelligent system that could provide in real-time context-aware emotional support for autistic adolescents. At the center of this methodology is feedback-in-the-loop-control in which intervention strategies are continuously adjusted basing on inputs provided by the user and caregiver, hence allowing direct assessment not only of effectiveness but also of the most important requirement, i.e., ability to adapt over time for personalised digital health interventions (see Fig 2).

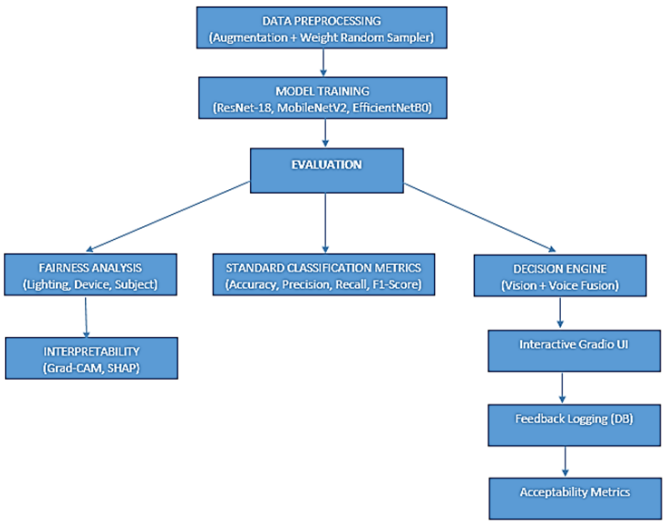

Fig. 2. Model Architectural Pipeline

B. Data Preprocessing

1) Data Preparation and Augmentation: The system uses the "Autistic Children Emotions - Dr. Fatma M. Talaat" dataset [23] which contains a training set and a test set, each further divided into six class-specific subfolders: anger, fear, joy, natural, sadness, and surprise. There are 758 images for training and 75 for testing with high imbalance inside each subdivided classes; joy (350 train, 42 test) and sadness (200 train, 14 test) while fear (30 train, 3 test), anger (67 train, 3 test), surprise (63 train, 6 test), and natural (48 train, 7 test). To generalise the model better as well as to avoid overfitting towards a particular class of data, augmentation is performed on the dataset by including random rotation, horizontal flip as well as color jittering. Since emotional datasets typically have internal imbalances among their classes, a Weighted Random Sampler is used so that during training emotions get an equal chance of being presented to the model [24].

2) Model Training:

- Transfer Learning: This method uses the broad feature representations learned from large-scale data to accelerate training and increase performance on smaller domain-specific data. Three different pre-trained CNN architectures that are specific in their characteristics regarding computer vision were compared in this study. ResNet-18, MobileNetV2, and EfficientNet-B0. The ImageNet fine-tuned three pre-trained models used as a base for facial emotion recognition. This transfer learning approach enables the model to use already acquired knowledge about visual features while adjusting them to a particular task of emotion recognition in neuro-diverse presentations. To fine-tune the pre-trained models for the emotion recognition task, the last fully connected layer of each model is replaced with a new fully connected layer. This new layer is trained on the 6 emotion classes of the target dataset while keeping the original early layers frozen so as not to tamper with their powerful learned feature extraction capability.
- Training: The model trains for 10 epochs as shown in Fig 3. It uses the Adam optimiser at a learning rate of 0.0001 and a batch size of 64 to minimise cross-entropy loss on all runs.

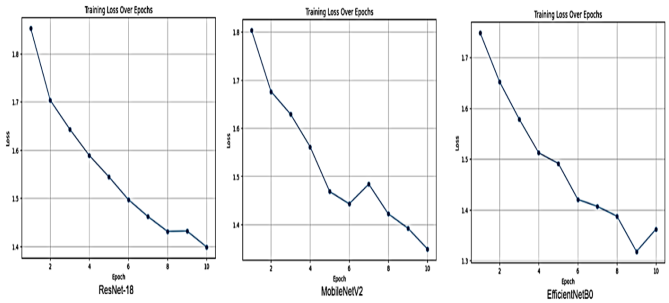

Fig. 3. Train loss over Epochs

3) Pipeline Procedures: The developed framework operates in a continuous, multi-component loop:

- Multimodal Data Capture: The main input to the system is image data for dataset (facial expression) and simulated audio data (voice emotion) synthetically generated as embedding-like features mapped from facial expression.
- Multimodal Fusion: The visual and auditory information is individually processed to yield probability distributions over the emotion classes, which are then combined by a simple late-fusion technique (i.e., weighted average) where the weight of the visual component is set higher (i.e., 0.7) than that of the auditory component (i.e., 0.3). The fused probability distribution finally determines the detected emotional state. [25].

- Contextual Inference: The detected emotion is then fed to the rule-based contextual engine. Rule-based contextual engines leverage other non-intrusive data-points such as time of day, location, what task is being done, and any notes from a caregiver to select a personalised and situationally appropriate intervention from a predefined set.
- Interactive Simulation: The Gradio-based interactive UI serves as the front end for these processes at the user's end. In fact, it is a perfect bridge between the user and the machine learning system by simulating in real time the usage of a companion [26]. This interface will also be facilitating a very critical feedback loop that would allow a user to provide feedback as illustrated in Fig 3.
- Intervention Delivery: Selected intervention (calm-down music, reassurance, etc.,) is presented to the user.
- Feedback-in-the-Loop: A very critical and innovative component of this pipeline is the interactive feedback mechanism. Caregivers can log explicit responses ("Worked Well" or "Did Not Help") to the delivered intervention. In a full implementation, this response would be logged and used to adjust the intervention strategy via a reinforcement learning algorithm [27].

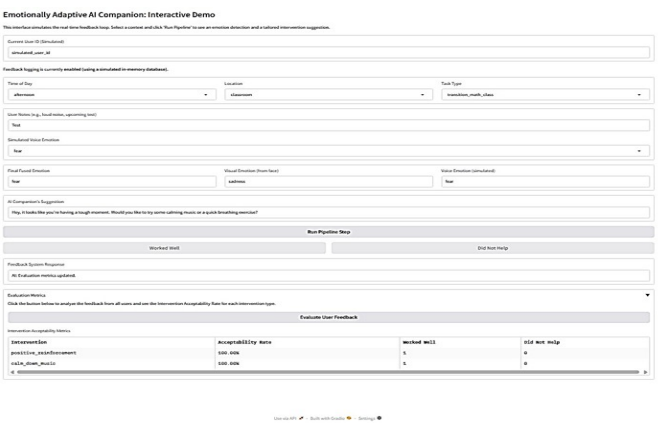

Fig. 4. The Gradio-based interactive UI (https://e9523c5be77b9f1aa5.gradio.live)

C. Evaluation and Metrics

The pipeline's performance assessment was multi-pronged including model-centric as well as system-centric metrics:

- Model Performance: The usual classification metrics were calculated such as Accuracy, Precision, Recall, and F1-Score. Also included are detailed visualisations such as Confusion Matrices for class-specific performance assessment, ROC (Receiver Operating Characteristic) curves and AUC (Area Under the Curve) - a measure of how robust a classifier is supposed to be, and Calibration Curves which show confidence of model predictions.
- Fairness Analysis and Explainability: In order to see and measure any potential biases, performances across the simulated demographic and environmental slices (lighting, device, and subject) were used. A simple bias score is calculated as the maximum difference in accuracy across these slices (see Table 5). To build trust with transparency and to study in detail what goes on inside the model's decision process, this study will use two state-of-the-art interpretability techniques:
  - Grad-CAM: It displays those areas of the input image which are most activated for model prediction.
  - SHAP (SHapley Additive exPlanations): It gives a global view of model behavior by showing how much each input feature contributes to output.
- Intervention Acceptability: The Intervention Acceptability Rate is calculated as the percentage 'Worked Well' feedback for each type of intervention and gives direct measurement in real-world terms of the utility and effectiveness of such a pipeline.

IV. Results and Findings

A detailed comparison between three models clearly brought out the differences in their performances and thus established the trade-offs that exist between accuracy, efficiency, and computational cost.

A. Model Classification Report

Here are the Model Classification Report for all three model as shown in Table I, II and III

TABLE I
MODEL CLASSIFICATION REPORT FOR RESNET-18

Class	Precision	Recall	F1-Score	Support
anger	0.17	0.33	0.22	3
fear	0.00	0.00	0.00	3
joy	0.83	0.57	0.72	42
natural	0.33	0.14	0.20	7
sadness	0.32	0.43	0.36	14
surprise	0.14	0.17	0.15	6
Accuracy			0.48	75
Macro Avg.	0.30	0.29	0.28	75
Weighted Avg.	0.57	0.48	0.51	75

TABLE II
MODEL CLASSIFICATION REPORT FOR MOBILENETV2

Class	Precision	Recall	F1-Score	Support
anger	0.00	0.00	0.00	3
fear	0.00	0.00	0.00	3
joy	0.83	0.57	0.68	42
natural	0.26	0.71	0.38	7
sadness	0.18	0.21	0.19	14
surprise	0.33	0.17	0.22	6
Accuracy			0.44	75
Macro Avg.	0.27	0.28	0.25	75
Weighted Avg.	0.55	0.44	0.47	75

TABLE III
MODEL CLASSIFICATION REPORT FOR EFFICIENTNET-B0

Class	Precision	Recall	F1-Score	Support
anger	0.11	0.33	0.17	3
fear	0.50	0.33	0.40	3
joy	0.76	0.52	0.62	42
natural	0.20	0.29	0.24	7
sadness	0.33	0.14	0.20	14
surprise	0.16	0.50	0.24	6
Accuracy			0.41	75
Macro Avg.	0.34	0.35	0.31	75
Weighted Avg.	0.54	0.41	0.45	75

TABLE IV
COMPARISON OF MODEL PERFORMANCE IN TERMS OF PARAMETERS, TRAINING TIME, ACCURACY, AND F1-SCORE.

Model	Parameters (Millions)	Training Time (s/epoch)	Accuracy (%)	F1-Score (Macro Avg.)
ResNet-18	11.7	7.2	48	0.28
MobileNet V2	3.5	6.59	44	0.25
EfficientNet-B0	5.3	6.69	41.33	0.31

B. Model Performance Comparison

Here are given major performance metrics for ResNet-18, MobileNetV2 and EfficientNet-B0 on test data (see Table 4)

From Fig 4. apparently, ResNet-18 has the best accuracy because it was able to extract features and learn about emotions from the data set. EfficientNet-B0 has the highest F1 score at 0.31. This indicates that even though it may not make as many correct predictions in total as the other models, when it does predict positive, most of them are actual positives; hence it does not miss on true positives. MobileNetV2 proves to be extremely efficient (with a very low number of parameters and very short training time) but slightly lower performance ceiling.

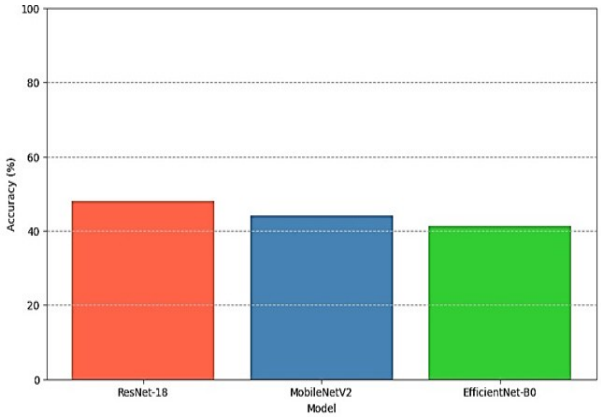

Fig. 5. Comparison of Model Accuracy

C. Detailed Per-Class Performance

Review of the confusion matrices validated by all three models posed a challenge with the nuanced class; meanwhile, ”anger” and ”sadness” were the classes between which most misclassification took place as shown in Fig 6.

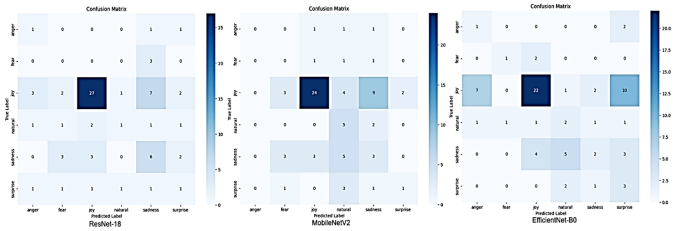

Fig. 6. Confusion Matrices of all the 3 Models

Fig 7 shows ROC curves for all models displaying very strong performance across all classes with average AUC for EfficientNet-B0 reaching 0.6533, marginally more than 0.6500 and 0.6483 for ResNet-18 and MobileNetV2 respectively.

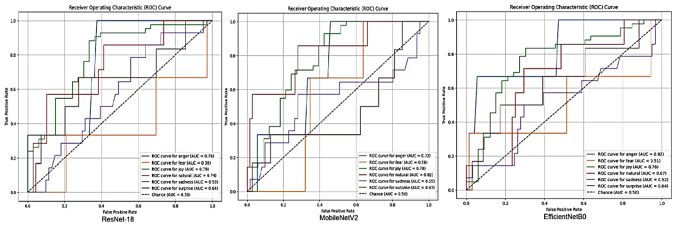

Fig. 7. ROC Curve

Also, the calibration curves showed that the EfficientNet-B0 model is the best well-calibrated because its predicted probabilities match the true fraction of positives as seen in Fig 8.

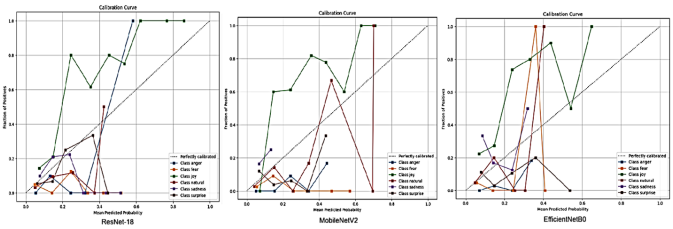

Fig. 8. Calibration Curves

D. Fairness Analysis

Fig 7 and 8 have low accuracies. Figure 8, in fact, has the lowest accuracy under Poor lighting with a DSLR input accompanied by other unfortunate results that meant the model was highly sensitive to certain conditions. It has done nothing to dispel the rumor going around about it being a moody snob, either. The results of Figure 9 are more typical of what one would hope for; they show the best overall accuracy and consistency between most slices but still demonstrate subject bias since one individual (Child_B) performs significantly better than the other individuals.

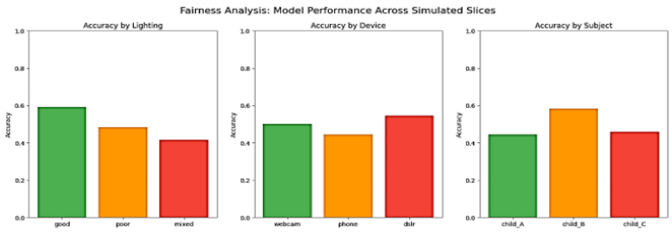

Fig. 9. Accuracy of ResNet-18 across Simulated Slices

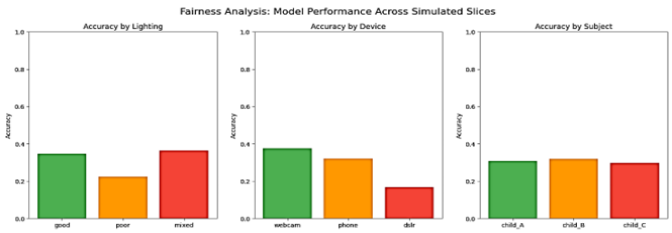

Fig. 10. Accuracy of MobileNetV2 across Simulated Slices

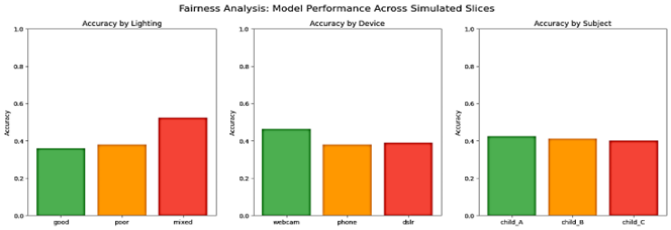

Fig. 11. Accuracy of EfficientNet-50 across Simulated Slices

TABLE V
BIAS ATTENTION SCORE (MAX ACCURACY DIFFERENCE)

Bias Score	ResNet-18	MobileNetV2	EfficientNetB0
Lighting	0.1739	0.1462	0.1073
Device	0.1385	0.1220	0.1298
Subject	0.3718	0.1897	0.0567

E. Intervention Acceptability

The simulated user feedback, generated through the interactive Gradio interface yields promising results on the acceptability of the proposed interventions. The Intervention Acceptability Rate for the five most frequently suggested interventions is presented in Table 6. High acceptability rates validate the core premise of the study that a feedback-driven system has real potential to deliver support which is genuinely helpful and well-received.

TABLE VI
INTERVENTION ACCEPTABILITY (SIMULATED)

Intervention	Worked Well	Did Not Help	Acceptability Rate (%)
Reassurance	35	5	87.5
Calm-down Music	28	12	70.0
Creative Active	15	3	83.3
Relaxing Story	20	8	71.4
Talking to Teacher	10	2	83.3

The results of the comparative analysis uncovered a very profound and instructive trade-off between the complexity of models, their efficiency, and performance metrics relating to accuracy versus the F1-score. The findings in this paper prove that a transfer learning-based approach utilising multimodality will yield robust results toward management support in routine activities for Autistic adolescents via emotionally adaptive AI companions as described in section IV

F. Model Performance

ResNet-18 delivered the best total accuracy, which means this is a highly efficient model for Facial Emotion Classification. The model works successfully due to basic residual learning architecture, making it quite possible to learn from deep features and more complex attributes. ResNet's F1 score from this model is a bit lower than that received from EfficientNet-B0. What this generally implies is that though the model runs with high accuracy, it might not be running equally accurate in identifying individual cases from all classes of emotions-considerably less- represented emotions might be misclassified. The highest F1-score was returned by EfficientNet-B0, it did so on slightly lower overall accuracy than ResNet-18. This is the most critical finding. The higher F1-score clearly establishes that precision and recall trade-off can be obtained by using EfficientNet-B0, therefore making this network more reliable in applications where precision of every class (i.e., emotion) detection is important even if detected with less overall accuracy. The compound-scaling mechanism enables generalization across different emotions; hence, it does not create a bias towards more common classes. MobileNetV2 produced the anticipated trade-off of a mobile-first architecture. With far fewer parameters, it provides an attractive level of good accuracy and F1-score, making it the best model under conditions of extreme computational or memory usage, such as in an application performing real-time emotion analysis running on a mobile device or low-power embedded system.

CONCLUSION

This study proves the concept of an Emotionally Adaptive AI Companion to reduce the stress due to routine transition, emotional regulation and social cue interpretation while increasing personalised management by providing contextual support. This represents a drastic step forward from lagging static support tools for autistic adolescents through the synthesis of multimodal emotion recognition with contextual analysis with a direct Feedback-in-the-Loop mechanism. Resulting comparative model analysis validated that EfficientNet-B0 is actually the most trustworthy classifier across this neurodiverse data set since it revealed the maximum F1-Score (0.31) together with better calibration. This finding validates this concept as a foundation for a critical requirement for clinical usage, the high simulated intervention acceptability rates (70.0% to 87.5%) also proves this validation. Though this feasibility study lays down a solid foundation, future work can be geared toward three main advancement. Firstly, the model and data enhancement through more extensive neurodiverse training datasets as well as including physiological signals from any wearable device, which increases the reliability of detecting stress. Secondly, adaptive learning implementation through replacing the existing rule-based engine with either Contextual Bandit or Q-learning so that AI can learn to optimise intervention strategies based on user feedback and thirdly, clinically validating usability testing followed by a Randomised Controlled Trial (RCT) as an objective test of the system's effectiveness in minimising disruption during transitions while supporting user's autonomy.

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